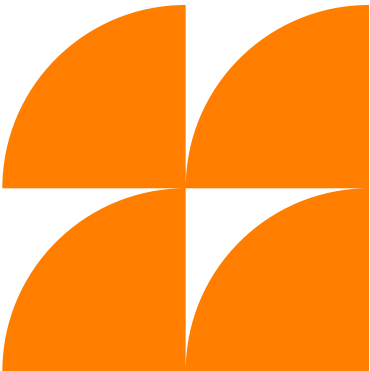




Problem Statement

Proposed Solution

Develop a Video Search System that allows users to search video content using text-based queries and directly jump to the relevant timestamp in the video.

When a user enters a keyword, the system searches the transcription to find where that word appears and displays the corresponding timestamps. The user can then directly play the video from the selected timestamp, allowing quick access to the required part of the video.

Objectives of the Project

The main objectives of the project are:

1) Convert video speech into text using speech recognition.

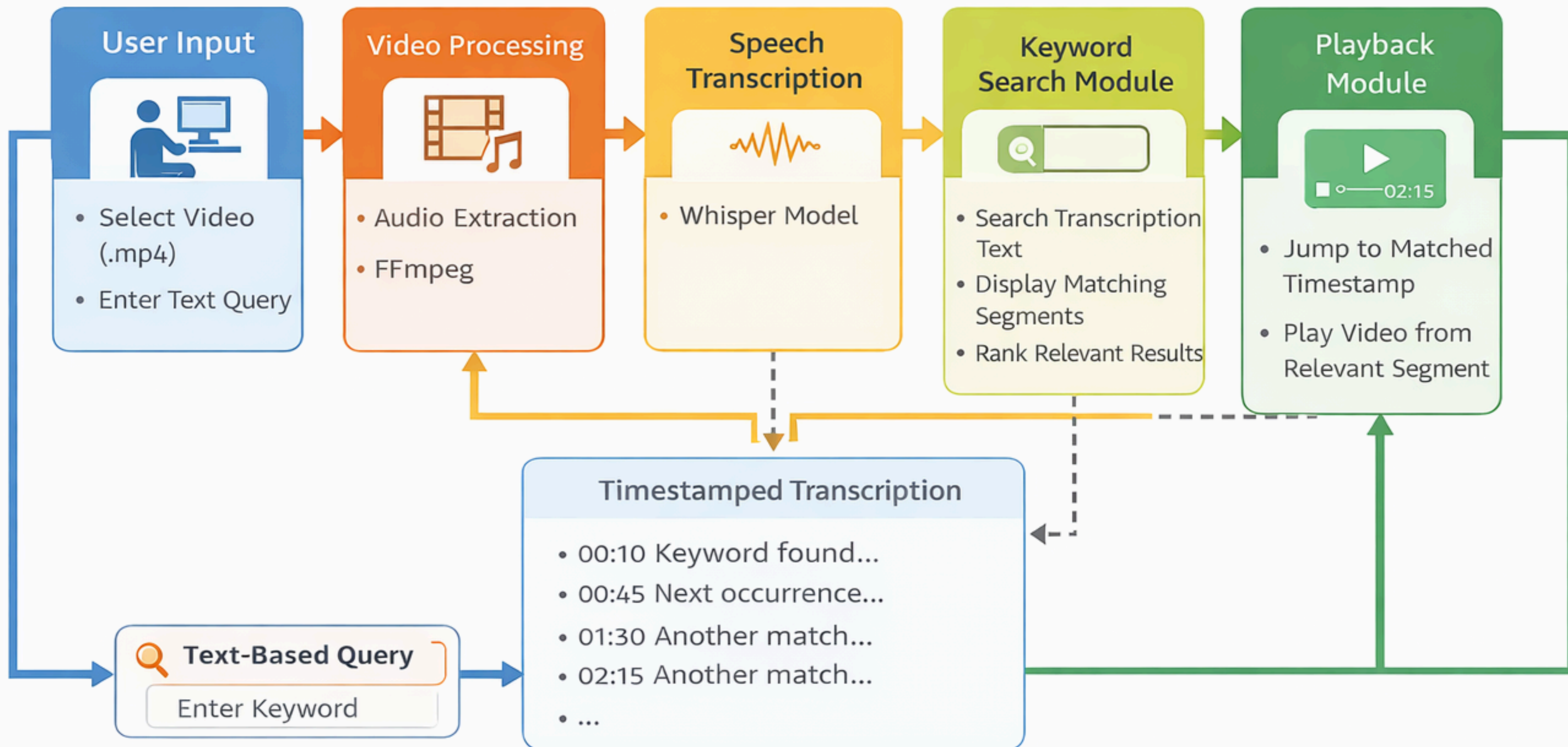
2) Allow users to search keywords within video content.

3) Retrieve relevant video segments quickly.

4) Enable direct playback from the matched timestamp

5) The system aims to improve video content accessibility and search efficiency.

System Architecture



Methodology / Working Process

01

User selects a video file (.mp4).

05

User enters a keyword query

02

Audio is extracted using FFmpeg.

06

System searches keyword in transcription text

03

Whisper model transcribes speech into text.

07

Matching segments are ranked and displayed

04

Transcription is divided into timestamped segments.

08

Video can be played directly from the selected timestamp

Modules Implemented So Far

1. Video Processing Module	2. Speech Transcription Module	3. GUI Interface
• Extracts audio from video files using FFmpeg.	• Converts audio into text using Whisper. • Generates timestamped speech segments.	Developed using Tkinter with features: • Select video • Process video • Enter search keyword • Display search results • Play video from timestamp

Current Project Progress

Week 1

- Problem identification
- Requirement analysis
- System architecture design

Week 2

- Literature survey
- Study of speech recognition models
- Design of transcription pipeline

Week 3

- Implementation of video processing pipeline
- Timestamp-based transcription using Whisper
- GUI development using Tkinter
- Keyword search functionality implemented

Future Work

Upcoming Development :

- Faster keyword search algorithms.
- Improved ranking mechanism.
- Progress bar for video processing.
- Display timestamps in **HH:MM:SS** format.
- Integrate video playback inside the application.
- Testing on multiple video datasets and languages.

Expected Outcome

A system that allows users to search videos using text queries and directly access relevant content segments.



Thank you!